Kidney-RAG

Clinical Retrieval-Augmented Generation over CKD Guidelines

Project Documentation

Creativa × Orange AI Hackathon — August 2026

Project	Kidney-RAG
Domain	Clinical decision support / nephrology
Corpus	KDIGO 2024, NICE NG203, KDIGO 2022 Diabetes, USPSTF
Chunks	666 (median 510 tokens, hard cap 750)
Embeddings	abhinand/MedEmbed-large-v0.1 (1024-dim)
Vector DB	ChromaDB (cosine, persistent)
Retrieval	Hybrid RRF (0.7 semantic / 0.3 BM25), top-k=5
Hit@5	0.867 (semantic & hybrid)
Faithfulness	1.0 on Day-4 eval set

Contents

1. Executive Summary

Kidney-RAG is a clinical Retrieval-Augmented Generation system over official public Chronic Kidney Disease (CKD) guidelines. It answers clinician-style questions by retrieving grounded evidence from four authoritative sources (KDIGO 2024, NICE NG203, KDIGO 2022 Diabetes, USPSTF) and generating answers in which every clinical claim traces back to *document* + *section* + *page* + *chunk_id*. The guiding principle is **Fluent Answer ≠ Safe Answer**: the system prefers refusal over fabrication and explicitly labels evidence strength.

The pipeline was built and evaluated over five days. Day 1–2 produced a searchable vector database with an evaluated hybrid retriever (Hit@5 = 0.867). Day 3 added grounded generation with mandatory citations. Day 4 added a safety layer (claim extraction, faithfulness scoring, citation accuracy, 4-level uncertainty language) and a web UI. Day 5 focuses on deployment and demo.

2. Problem & Motivation

General-purpose LLMs hallucinate confidently on medical questions, cite non-existent papers, and give no way for a clinician to verify a claim. This is not a fine-tuning problem — it is an evidence-retrieval and attribution problem. Kidney-RAG constrains the model to a fixed, versioned corpus of official guidelines and forces every clinical claim to carry a traceable citation. Questions outside the corpus (e.g. dialysis dosing, pediatric CKD, transplant surgery) trigger an explicit refusal rather than a plausible-sounding invention.

3. System Architecture

The system is a linear ingestion pipeline feeding an online hybrid retriever, a grounded generator, and a post-hoc safety verifier. All components share the frozen chunk contract described in section 5.

Pipeline overview

```
raw_pdfs/ pdf_parser.ipynb chunker.ipynb embedder.ipynb retrieval.py
KDIGO --> parsed/kdigo.json --> | |--> HybridRetriever
NICE --> parsed/nice.json --> chunks/ |---> embeddings.npy
KDIGO-DM --> parsed/kdigo_dm.json --> all_chunks.jsonl | chroma_db/ |
USPSTF --> parsed/uspstf.json --> | (ckd_guidelines) |
v
generation.py <-- retrieval --> safety.py --> web/backend/app.py
```

Request lifecycle

• User submits a clinical question via the web UI (POST /api/ask).

• HybridRetriever prefixes the query, embeds it with MedEmbed, and fetches top-50 semantic candidates from Chroma.

• BM25 runs over the same 666-chunk corpus and returns lexical candidates.

• Weighted Reciprocal Rank Fusion (0.7 semantic / 0.3 BM25, k=60) fuses the two ranked lists and keeps the top-5.

• KidneyRAGGenerator formats the 5 chunks into a strict citation prompt and calls the LLM (Groq / Gemini pool).

• safety.py extracts atomic clinical claims, verifies each against the retrieved evidence, computes faithfulness and citation accuracy, and maps the aggregate to a 4-level evidence-strength label.

• The API returns the answer, the ordered citations, faithfulness/citation metrics, evidence-strength label, and any refusal flag.

4. Corpus & Ingestion

Document	Pages	Role
KDIGO 2024 CKD Guideline (full)	199	Primary CKD management
NICE NG203 CKD	78	UK diagnosis / pathway
KDIGO 2022 Diabetes Management in CKD	128	SGLT2 / GLP-1 / finerenone
USPSTF CKD Screening	6	Safe-refusal I-statement

Parser choice: PyMuPDF (*fitz*). pdfplumber was tested and rejected because it glued words together on KDIGO's two-column layout. Two PDF-specific quirks are handled during parsing:

• **Character corruption:** KDIGO PDFs render '≥' as '$' (67 places) or dagger (25 places). The parser normalises these contextually — only when adjacent to a digit — so legitimate dollar signs and footnote daggers are preserved.

• **Ligatures:** Unicode ligatures (fi, fl, ff, ffi, ffl) are normalised to their two-letter equivalents to prevent tokenisation misses at retrieval time.

• **NICE footer noise:** the footer spans three separate lines (document title, 'Page X', 'of 78') and each is stripped individually.

The KDIGO Summary of Recommendations and Executive Summary PDFs were held out (they duplicate the body) and used as the answer key when building the eval set.

5. Chunking Strategy

Chunks are the atomic unit of retrieval and citation. The contract is frozen — every downstream component depends on it.

```
{"chunk_id": "kdigo_dm_p44_c02",
 "document_name": "KDIGO 2022 Diabetes Management in CKD",
 "source_url": "https://kdigo.org/...",
 "page_number": 44,
 "page_range": [44, 48],
 "section_title": "1.3 Sodium-glucose cotransporter-2 inhibitors",
 "token_count": 483,
 "text": "..."}
```

Key rules

• **ID format:** <doc-short>_p<start_page>_c<index> where doc-short is kdigo, nice, kdigo_dm, or uspstf.

• **Size:** 550 tokens with 55-token overlap. Hard cap 750. Median 510. Zero oversized chunks.

• **Total:** 666 chunks — KDIGO 340, NICE 72, KDIGO-DM 241, USPSTF 13.

• **Section filter:** is_clinical_section() drops research agendas, author bios, notices, and acknowledgments.

• **Skip pages:** KDIGO 2024 pp 34-53 and KDIGO 2022 pp 20-29 are excluded because they are Summary-of-Recs duplicates of the body.

• **Chroma metadata:** flat scalars only. page_range is split into page_start / page_end integers.

• **Adding a document:** one config dict in pdf_parser cell 3 + one line in chunker PARSED_FILES + one line in SKIP_PAGES. embedder needs zero changes.

6. Embeddings & Vector Index

The embedding model is **abhinand/MedEmbed-large-v0.1**, a 1024-dim BGE-large variant fine-tuned on PubMed. It is loaded locally via sentence-transformers — no API, no rate limits. It beat BAAI/bge-large-en-v1.5 by **+14 points Hit@5** on our eval set (0.867 vs 0.733).

Queries are prefixed with the bge-style instruction *"Represent this medical question for retrieving relevant clinical guideline passages: "*. All vectors are L2-normalised so cosine similarity equals dot product.

Vectors are persisted in ChromaDB (cosine space, collection *ckd_guidelines*) at *corpus/chunks/chroma_db/*. A parallel NumPy cache (*chunk_embeddings.npy*) speeds up model_compare and ablation scripts; the embedder reuses it if the shape matches (n_chunks, 1024).

7. Hybrid Retrieval

HybridRetriever.hybrid_search(query, k) runs semantic and lexical retrieval in parallel and fuses them with weighted Reciprocal Rank Fusion (RRF).

• **Semantic:** MedEmbed query vector against the Chroma cosine index, top-50 candidates.

• **Lexical:** BM25 (rank_bm25) over the same 666-chunk corpus, top-50 candidates.

• **Fusion:** weighted RRF, weights 0.7 semantic / 0.3 lexical, damping k=60.

• **Top-k = 5:** chosen because Hit@k plateaus at k=5 on the eval set.

• **Why hybrid:** ties semantic-only on Hit@5 (both 0.867) but is measurably more robust on threshold-anchored queries (e.g. 'ACR >300 mg/g').

8. Grounded Generation

The generator (*generation.py :: KidneyRAGGenerator*) receives the 5 top chunks and constructs a strict prompt that requires every clinical claim to end with an inline citation of the form *[chunk_id, document, page]*. Uncited clinical claims are treated as hallucinations by the downstream safety layer.

Backends: a Groq-hosted Llama model is the primary backend (fast, high citation accuracy in Day-4 evaluation). Google Gemini is a fallback via a multi-key pool. Keys are loaded via python-dotenv from *.env* (gitignored).

9. Safety & Verification Layer

safety.py is applied post-generation. It turns a fluent answer into a verified answer.

• **Claim extraction:** the answer is decomposed into atomic clinical claims.

• **Verification:** each claim is checked against retrieved evidence via three configurable verifiers — LLM entailment, embedding similarity, and NLI (natural-language inference).

• **Faithfulness score:** fraction of claims supported by retrieved evidence. Day-4 eval: 1.0.

• **Citation accuracy:** fraction of inline citations that resolve to a real returned chunk_id. Day-4 eval: 1.0.

• **Evidence-strength label:** 4-level mapping (Strong / Moderate / Limited / Insufficient) driven by retrieval scores, verifier agreement, and citation density.

• **Refusal:** if the out-of-scope score exceeds the tuned cutoff (~0.70 max cosine, calibrated against the OOS probe set), the system refuses with an explicit 'outside corpus scope' message rather than answering.

10. Evaluation Results

Metric	Value	Source
Semantic Hit@5	0.867	artifacts/day2/evaluation.csv
Hybrid Hit@5 (0.7 / 0.3 RRF)	0.867	artifacts/day2/evaluation.csv
BM25 Hit@5	0.733	artifacts/day2/evaluation.csv
MedEmbed vs BGE-large (Hit@5)	0.867 vs 0.733	artifacts/day2/model_compare.json
Chunk ablation winner	400/40 (Hit@5 0.933 page-level)	artifacts/day2/chunk_ablation.json
Top-k elbow	k=5	artifacts/day2/topk_curve.csv
OOS max cosine	0.663 (refusal cutoff ~0.70)	artifacts/day2/out_of_scope.json
Faithfulness (Day-4 eval)	1.0	artifacts/day4/summary.json
Citation accuracy (Day-4 eval)	1.0	artifacts/day4/summary.json

Eval set: **18 labelled questions** at *eval/eval_set.json* with human-verified gold chunk_ids. Includes strong-answer cases, partial-answer cases, and out-of-scope refusals (Case C, USPSTF).

11. Web Application

The demo UI is served by FastAPI with a self-contained static frontend (no external CSS/JS dependencies). Dark theme, mobile-responsive.

Endpoints

• **GET /health** — liveness probe.

• **GET /api/sources** — list indexed documents (name, pages, chunk count).

• **POST /api/ask** — body {question}. Returns {answer, citations[], faithfulness, citation_accuracy, evidence_strength, refused}.

• **GET /docs** — auto-generated Swagger UI.

Run locally

```
python -m uvicorn web.backend.app:app --port 8000
then open http://127.0.0.1:8000
```

12. Repository Layout

```
Kidney-RAG/
|-- corpus/
| |-- raw_pdfs/ # 4 source PDFs (committed)
| |-- parsed/*.json # parser output
| |-- chunks/all_chunks.jsonl # 666 chunks, source of truth
| |-- chunks/chroma_db/ # gitignored, regenerable
| |-- manifest.json, SOURCES.md
|
|-- pdf_parser.ipynb # PyMuPDF parsing
|-- chunker.ipynb # 550/55 windowing + section filter
|-- embedder.ipynb # MedEmbed + Chroma persist
|-- retrieval.py # HybridRetriever (semantic + BM25 + RRF)
|-- generation.py # KidneyRAGGenerator (grounded prompt)
|-- safety.py # claim extraction, faithfulness, refusal
|-- evaluate.py # Precision@k / Recall@k / Hit@k sweep
|-- evaluate_day4.py # full pipeline harness -> artifacts/day4/*
|-- model_compare.py, ablation.py, explain.py
|-- test_safety.py # 18 offline tests, no API needed
|
|-- eval/eval_set.json # 18 labelled questions
|-- artifacts/day2/ # eval outputs (JSON + CSV)
|-- artifacts/day4/ # safety eval outputs
|
|-- web/backend/app.py # FastAPI
|-- web/frontend/ # static UI
|
|-- docs/ # decks, teammate guides, this PDF
|-- models/ # gitignored embedding caches
|-- .env # gitignored, holds API keys
|-- CLAUDE.md # project brief
```

13. How to Run

Ingestion (only after corpus changes)

```
python -m jupyter nbconvert --to notebook --execute pdf_parser.ipynb --inplace
python -m jupyter nbconvert --to notebook --execute chunker.ipynb --inplace
python -m jupyter nbconvert --to notebook --execute embedder.ipynb --inplace # ~15-20
min first run
```

Evaluation & retrieval sandbox (assumes the index exists)

```
python evaluate.py --sweep-k # Precision@k, Recall@k, Hit@k, OOS max cosine
python model_compare.py # MedEmbed vs BGE-large
python ablation.py # chunk-size ablation
python explain.py "your clinical question"
```

Day-4 safety evaluation

```
python evaluate_day4.py --skip-generation # retrieval-only, deterministic
python evaluate_day4.py --verifier llm # full run, needs LLM quota
```

Web demo

```
python -m uvicorn web.backend.app:app --port 8000
```

14. Design Decisions & Trade-offs

• **Local embeddings over hosted API:** MedEmbed-large runs on-device, so retrieval is free, deterministic across runs, and never rate-limited during the hackathon demo.

• **Hybrid RRF instead of pure semantic:** ties on Hit@5 but is more robust on threshold-anchored queries where BM25 excels (numeric ranges, drug-name lookups).

• **PyMuPDF over pdfplumber:** pdfplumber glued adjacent tokens on KDIGO's two-column layout, which broke retrieval word-by-word. PyMuPDF preserves separators reliably.

• **Frozen chunk contract:** every downstream tool joins on chunk_id. A schema change would invalidate the eval set, the citation format, and the Chroma index.

• **4-level evidence strength instead of a raw score:** clinicians read labels faster than probabilities. The mapping is calibrated from retrieval score, verifier agreement, and citation density.

• **Explicit refusal:** when the closest retrieved chunk is below the OOS cutoff, the system refuses. Fluency without a grounded citation is treated as a bug, not a feature.

• **No fine-tuning, no fine-tuning-time training:** the entire system is prompt + retrieval + verification. Swapping the LLM backend is a one-line change; swapping the embedding model requires a full re-embed of chunks and queries.

15. Known Limitations & Future Work

• **Corpus scope:** adult CKD management only. Pediatric CKD, dialysis prescription, transplant surgery, and non-CKD nephrology are all out of scope and correctly refused, but users may expect them.

• **Chunk ablation:** 400/40 outperformed the shipped 550/55 on page-level Hit@5 (0.933 vs 0.867), but the ablation used a different embedding model and coarser scoring. Chunk size is a Day-3+ re-validation candidate.

• **Verifier ensemble:** the LLM verifier depends on external quota. The similarity+NLI fallback is more robust but slightly less precise on paraphrased claims.

• **UI:** single-turn only. No conversation memory, no comparison view between guidelines. Deliberate scope choice for the demo.

• **Deployment:** Day-5 deliverable. Target hosts are Render or Railway; the FastAPI + static frontend layout is deploy-ready as a single service.

16. Team & Timeline

#	Role	Day 1-2	Day 3-5
P1	KDIGO owner	Parse + chunk KDIGO	Grounded generation
P2	NICE owner	Parse + chunk NICE	Grounded generation
P3	Index / retrieval	Embedder + Chroma + hybrid + eval	Retrieval tuning
P4	Eval / safety	Eval set + questions + QC	Safety layer + demo

Delivered milestones

• **Day 1** — Searchable Vector DB with Metadata. Retrieval verified end-to-end.

• **Day 2** — Retrieval Optimization. All 6 deck items done, every claim traces to artifacts/day2/*.

• **Day 3** — Grounded generation + citation. HybridRetriever feeds KidneyRAGGenerator.

• **Day 4** — Safety, verification, evaluation, web UI. Faithfulness 1.0, citation accuracy 1.0.

• **Day 5** — Deploy web UI, rehearse 3-question demo (strong / partial / refusal), final deck.

Kidney-RAG — Creativa × Orange AI Hackathon, August 2026.